

Lab Assignment - 1

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① Greedy algorithm for a multi-Armed problem

given

Machine A $\rightarrow [3, 2, 4]$

" B $\rightarrow [6, 7]$

" C $\rightarrow [5, 4, 6]$

formula :-

$$Q(a) = \frac{\text{sum of rewards}}{n(a)}$$

Machine A :

$$Q(A) = \frac{3+2+4}{3} = 3.0$$

Machine B :

$$Q(B) = \frac{6+7}{2} = 6.5$$

Machine C :

$$Q(C) = \frac{5+4+6}{3} = 5.0$$

The greedy option is B = 6.5

Q Greedy algorithm using incremental update method.

Machine (A) [3, 2, 4]

" (B) [6, 7]

" (C) [6, 4, 6]

Initially

$$Q(A) = 0 \quad N(A) = 0$$

$$Q(B) = 0 \quad N(B) = 0$$

$$Q(C) = 0 \quad N(C) = 0$$

formula:

$$Q_{\text{new}}(a) = Q_{\text{old}}(a) + \frac{R - Q_{\text{old}}(a)}{N(a)}$$

Machine A's Rewards: 3, 2, 4

first reward = 3

initially

$$Q(A) = 0$$

$$N(A) = 0$$

$$N(A) = 1$$

$$\text{update } Q(A) = 0$$

$$Q(A) = 3$$

Second $R = 2$

$$N(A) = 2$$

$$Q_{old}(A) = 3$$

$$Q(A) = 3 + \frac{2 - 3}{2}$$

$$Q(A) = 3 - \frac{1}{2}$$

$$Q(A) = 2$$

Third $R = 4$

$$N(A) = 3$$

$$Q(A) = 2.5 + \frac{4 - 2.5}{2}$$

$$Q(A) = 2.5 + \frac{1.5}{3}$$

$$Q(A) = 3.0$$

Machine B (Rewards) (6, 7)

~~First~~ $N(B) = 2$

$$Q(B) = 6.5$$

Machine -

~~first~~ $R = 5$

$$N(C) = 1$$

$$Q_{old}(C) = 0$$

$$Q_{new} = 5$$

Second $R = 4$

$$N(C) = 2$$

$$Q_{old}(C) = 5$$